

Testing automata web app

Hi everyone,

I'm a Level 3 Computer Science student, and for my final year project, I've developed a web app that visualises finite automata and related algorithms (NFA to DFA conversion, DFA minimisation, and regex to NFA conversion).

I'm looking for students to take part in a short user study to help evaluate how useful and effective the tool is. It would involve a brief refresher on the topic (if needed), using the tool for about 10 minutes, and then answering a few questions.

If you're interested, I'd really appreciate it - it would also be a great way to revise some key concepts!

Please reach out to me at sc21lh2@leeds.ac.uk if you'd like to participate or have any questions.

Thanks, Lindsey



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Formal Language & Finite Automata

Turing machines

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All reading is from the "Introduction to the theory of computation" - Micheal Sipser

- Turing machines - Pages 193 - 197 (essential)

Algorithm

Algorithm vs. Turing machine

Informal: Every sensible model of computation that allows arbitrary access to memory and each step of computation performs only a finite step of work will be equivalent to a Turing machine.

Informal: Everything an algorithm can compute can be computed by a Turing machine (Church-Turing thesis says that λ -calculus is equivalent to Turing machines).

Consequence: The formalization of the term algorithm allows us to prove that there are problems for which there is no algorithm (they are not Turing decidable).

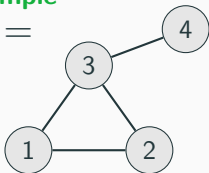
Encoding and simulation

Encoding

Given some object M , we denote its encoding as a string by $\langle M \rangle$.

Example

$G =$

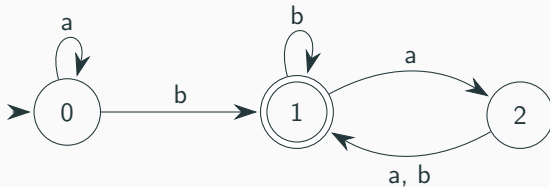


$\langle G \rangle =$

$(1, 2, 3, 4)((1, 2), (1, 3), (2, 3), (3, 4))$

Example

$M =$



$\langle M \rangle =$

$(0, 1, 2)(a, b)((0, a, 0), (0, b, 1), (1, a, 2), (1, b, 1), (2, a, 1), (2, b, 1))0(1)$

Simulation

Assume M is a machine (e.g. DFA, NFA, PDA or TM) and w an input string, a Turing machine can given $\langle M, w \rangle$ simulate the computation of M on w .

Example

Decidable languages

Languages related to regular languages

Define the following languages:

$$A_{\text{DFA}} = \{\langle A, w \rangle \mid A \text{ is a DFA that accepts } w\};$$

$$A_{\text{NFA}} = \{\langle B, w \rangle \mid B \text{ is an NFA that accepts } w\};$$

$$E_{\text{DFA}} = \{\langle A \rangle \mid A \text{ is a DFA and } L(A) = \emptyset\};$$

$$EQ_{\text{DFA}} = \{\langle A, B \rangle \mid A, B \text{ are DFA's and } L(A) = L(B)\}.$$

Theorem

The languages A_{DFA} , A_{NFA} , E_{DFA} and EQ_{DFA} are Turing decidable.